



Wasla

Transport Analytics & Operations Intelligence

Data-Driven Insights for Tunisian Intercity Transport

Internship Project Brief

Organisation	Backend Glitch Inc. (backendglitch.com)
Project	Wasla - Transport Analytics Engine
Live Stations	Monastir Centre, Jammel-Monastir
Upcoming	Ksar Hlel (and further stations)
Document Type	Internship Project Brief
Year	2026 / 2027
Classification	Internal

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1 Context

Wasla is a live transport management system developed by Backend Glitch Inc., currently deployed at two intercity stations in the Monastir governorate: **Monastir Centre Station** (the hub) and **Jammel-Monastir Station**. A third station at **Ksar Hlel** is planned, with further expansion expected across the region.

The platform handles the full operational workflow: passenger bookings, vehicle dispatching (louages / shared taxis), day pass management, route scheduling, ticket printing, staff shift tracking, and real-time station monitoring.

The system has been running in production since April 2026. The current fleet comprises **113 active vehicles** (8-seat capacity each) serving **5 destination routes**: Ksar Hlel, Tunis, Sousse, Souassi, and Monastir. The data volume is growing daily as the platform scales across stations.

This internship focuses on building the **analytics layer** on top of this operational data - transforming raw transactional records into actionable business intelligence for station operators and management.

2 The Data

The intern will work directly with the live Wasla PostgreSQL database. Key tables and their current volumes:

Table	Status	Content
bookings	Active	Passenger bookings with status, amount, payment method, timestamps, destination
day_passes	Active	Station entry passes with validity windows
trips	Active	Vehicle departures with license plate, destination, seat count
vehicles	113 registered	Fleet registry (8-seat capacity per vehicle)
stations	2 live	Monastir Centre, Jammel-Monastir
routes	5	Ksar Hlel, Tunis, Sousse, Souassi, Monastir
staff	6	Station agents with shift tracking
staff_daily_statistics	Active	Per-agent transaction summaries and performance
station_daily_statistics	Active	Aggregated daily station metrics
print_jobs	Active	Ticket printing with audit trail

The data spans from April 2026 to present. All monetary values are in Tunisian Dinar (TND). The database grows daily with live station operations.

3 Project Pipeline

Stage		What Happens
Data Exploration	Explo- ration	Connect to the live database, explore schema and distributions, understand the business domain
		Data Cleaning Handle anomalies, normalise timestamps, engineer features (hour of day, day of week, seasonality markers)
Statistical Analysis	Anal- ysis	Passenger flow patterns, revenue trends, peak hour analysis, route performance comparison, fleet utilization
ML & Forecasting	Forecast- ing	Predict daily passenger volume, forecast revenue, detect anomalies in booking patterns
Dashboard	Dashboard	Build a real-time operations dashboard with key metrics, charts, and drill-down capability
Cloud Deployment	Deploy- ment	Deploy the analytics stack (API + dashboard) to a production VPS

3.1 Data Exploration & Cleaning

- Connect to the production PostgreSQL database and explore the schema
- Understand the business logic: how bookings flow, what each status means, how trips relate to vehicles and routes
- Profile the data: distributions, missing values, outliers
- Engineer time-based features: hour of day, day of week, month, season, holiday markers
- Create a clean, analytics-ready dataset for downstream work

3.2 Statistical Analysis

- **Passenger flow** - hourly and daily booking volume patterns, weekday vs weekend comparison, seasonal trends
- **Revenue analytics** - daily revenue trends, average revenue per booking, per-route revenue contribution
- **Peak hour analysis** - identify busiest hours per station per day, quantify peak-to-off-peak ratio
- **Route performance** - compare the 5 destinations by volume, revenue, and growth
- **Fleet utilization** - how many vehicles are active per day, trips per vehicle, average load factor
- **Cancellation analysis** - cancellation rate, reasons, timing patterns

3.3 Machine Learning & Forecasting

With cleaned and analysed data in hand, the next stage applies machine learning to extract predictive value from the historical booking patterns. Two models are trained, evaluated, and integrated into the dashboard:

3.4 Dashboard

Build a real-time operations dashboard using **Next.js**:

Module	What It Shows
Overview	Today's bookings, revenue, active vehicles, passengers per station
Passenger Flow	Hourly and daily booking trends, peak hour indicators
Revenue Dashboard	Daily/weekly/monthly revenue, per-route breakdown, growth trends
Fleet Monitor	Active vehicles, trips completed, load factor, vehicle utilization
Forecast View	Predicted vs actual volume, next 7-day forecast, anomaly flags
Station Drill-down	Per-station metrics: Monastir Centre vs Jammel-Monastir comparison

3.5 Cloud Deployment

Deploy to the same VPS infrastructure that runs Wasla:

- Containerise the analytics API (Python/FastAPI) and dashboard (Next.js) with Docker
- Configure Nginx reverse proxy for the analytics subdomain
- Connect to the live database (read-only replica or direct with read-only credentials)
- Set up automated data refresh (daily aggregation jobs)

4 Technology Stack

Component	Technology
Database	PostgreSQL (production, live operational data)
Data Processing	Python (pandas, numpy, SQLAlchemy)
Statistical Analysis	Python (matplotlib, seaborn, scipy)
Forecasting	Prophet / statsmodels / scikit-learn
Anomaly Detection	scikit-learn (IsolationForest, statistical thresholds)
Analytics API	Python (FastAPI)
Dashboard	Next.js (React) with Recharts/Chart.js
Containerisation	Docker + docker-compose
Reverse Proxy	Nginx
Hosting	Linux VPS (existing Wasla infrastructure)

5 Why This Project

- **Real data, real impact** - work with live production data from an operational transport system, not synthetic datasets
- **Full data science pipeline** - from raw SQL to deployed dashboard, every stage of the workflow
- **Clear deliverables** - each stage produces a tangible, demonstrable output
- **Business value** - the analytics dashboard will be used by actual station operators
- **Manageable scope** - the data model is clean and well-understood; no complex CV or blockchain
- **Portfolio piece** - a complete, deployed analytics system built on a live operational transport platform

6 Expected Deliverables

1. An exploratory data analysis report with visualisations (passenger flow, revenue, routes, fleet)
2. A cleaned and feature-engineered analytics dataset
3. A passenger volume forecasting model with documented accuracy metrics
4. An anomaly detection system with alert thresholds
5. A functional operations dashboard (Next.js) with real-time metrics
6. A Docker-deployed analytics stack running on the Wasla VPS
7. A final technical report documenting methodology, findings, and recommendations

7 Internship Timeline (Suggested)

Phase	Focus
Weeks 1-2	Database exploration, data cleaning, feature engineering, EDA report
Weeks 3-4	Statistical analysis: flow patterns, revenue, routes, fleet utilization
Weeks 5-6	ML models: forecasting + anomaly detection, model evaluation
Weeks 7-8	Dashboard development, cloud deployment, final report

This document is intended solely for the named internship candidate.

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